THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** 6	$2(x+10)=2x+3x+2 \Rightarrow x=6$
2	**(b)** Incorrect — it can be positive, negative or zero	e.g. $10, 7, 4, \ldots$ has $d=-3$
3	**(b)** 2	$a_n=S_n-S_{n-1}=2n+1 \Rightarrow d=2$
4	**(a)** $\frac{1}{3}, \frac{4}{3}$	$d=\frac{5}{3}-\frac{1}{3}=\frac{4}{3}$
5	**(b)** 80°	$\angle R=\angle C=180-100=80^\circ$
6	**(b)** 7.5 cm	$\frac{4}{6}=\frac{5}{EC}$
7	**(a)** Basic Proportionality Theorem	$\frac{AD}{DB}=1 \Rightarrow \frac{AE}{EC}=1$
8	**(b)** the same shape, not necessarily the same size	Definition of similarity
9	**(c)** −12	Mid-point $(-4,4) \Rightarrow \frac{a}{3}=-4$
10	**(a)** Student A is correct	$\sqrt{9+16}=5$; B added $3+4$
11	**(d)** $4\sqrt{2}$	$\sqrt{16+16}=4\sqrt{2}$
12	**(d)** 1 : 2	$\frac{6k-3}{k+1}=0 \Rightarrow k=\frac{1}{2}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$a=7,\ d=\frac{7}{2}$; $n=\frac{84-7}{3.5}+1=23$; $S=\frac{23}{2}(7+84)=1046\frac{1}{2}$	1+1
14	$5+6(n-1)=301 \Rightarrow n-1=\frac{296}{6}$, not a whole number ⇒ 301 is **not** a term	1+1
15	$\angle C=70^\circ, \angle Q=60^\circ$. So $\angle A=\angle P,\ \angle B=\angle Q,\ \angle C=\angle R \Rightarrow \triangle ABC \sim \triangle PQR$ (AA). Claim is false.	1+1
16	False. They are **similar** (SSS), not necessarily congruent; they are congruent only when the ratio is 1.	1+1
17	$\left(\frac{3\times8+4}{4}, \frac{3\times5-3}{4}\right)=(7,3)$	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$8, 16, \ldots, 120$: $S_{15}=\frac{15}{2}(8+120)=960$	1+1+1
19	$a=S_1=8$, $S_2=22 \Rightarrow a_2=14$, $d=6$; $a_n=8+6(n-1)=6n+2$	1+1+1
20	$EQ=1.10,\ FR=2.20$; $\frac{PE}{EQ}=\frac{0.18}{1.10}=\frac{9}{55}$, $\frac{PF}{FR}=\frac{0.36}{2.20}=\frac{9}{55}$ ⇒ equal ⇒ $EF \parallel QR$ (converse of BPT)	1+1+1
21	$\frac{2k+\frac{1}{2}}{k+1}=\frac{3}{4} \Rightarrow 8k+2=3k+3 \Rightarrow k=\frac{1}{5}$ ⇒ 1 : 5	1+1+1
22	$a+2d=4,\ a+8d=-8 \Rightarrow d=-2,\ a=8$; $8-2(n-1)=0 \Rightarrow n=5$: the 5th term	1+1+1
	OR: $5+(n-1)1.75=20.75 \Rightarrow n-1=9 \Rightarrow n=10$	1+1+1
23	$\frac{1.8}{1.2}=\frac{h}{24} \Rightarrow h=36$ m; shadow of pole $=\frac{9\times1.2}{1.8}=6$ m	2+1
	OR: $\angle ADB=\angle CEB=90^\circ$ and $\angle B$ common ⇒ $\triangle ABD \sim \triangle CBE$ (AA)	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	$7d=7 \Rightarrow d=1$; $S_{20}=10(6+19)=250$; $3+(n-1)=50 \Rightarrow n=48$ ⇒ yes, 48th term	1+2+2
	OR: $\frac{n}{2}[40-(n-1)]=200 \Rightarrow n^2-41n+400=0 \Rightarrow n=16$ or 25; $n=25$ gives negative top row, so 16 rows; top row $=20-15=5$ logs	3+2
25	(i) $A=(2\cdot2-1,\ 2(-3)-4)=(3,-10)$ (ii) $r=\sqrt{1+49}=5\sqrt{2}$ (iii) distance from centre $=7\neq5\sqrt{2}$ ⇒ does not lie on it	2+2+1
	OR: (i) $AC=\sqrt{185}$, $BD=\sqrt{164}$ (ii) Mid-points $(-3,0.5)$ and $(0,0)$ differ ⇒ diagonals do not bisect each other ⇒ not a parallelogram	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** 110	$5(4+18)=110$
2	**(d)** 34	$\frac{205-7}{6}+1=34$
3	**(c)** 3	$2(4k-6)=4k \Rightarrow k=3$
4	**(c)** Incorrect; the sum is n^2	$S_n=\frac{n}{2}[2+(n-1)2]=n^2$
5	**(d)** 6 cm	BPT: $\frac{2}{3}=\frac{4}{EC} \Rightarrow EC=6$
6	**(a)** 60 m	$\frac{12}{8}=\frac{h}{40}$
7	**(a)** 60°	$\angle X=\angle P=60^\circ$
8	**(b)** 36 cm	Perimeters are in the ratio of sides: $24\times\frac{3}{2}=36$
9	**(b)** $(5,4)$	$D=A+C-B=(5,4)$
10	**(d)** 13	$\sqrt{25+144}=13$
11	**(b)** $(3,5)$	$\left(\frac{8+1}{3},\frac{12+3}{3}\right)$
12	**(a)** $(2,6)$	$x+6=8,\ 4+y=10$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$5d=35 \Rightarrow d=7$, $a=38-70=-32$; $a_{31}=-32+30\times7=178$	1+1
14	$1,3,\ldots,49$: 25 terms; $S=\frac{25}{2}(1+49)=625$	1+1
15	If two angles of one triangle equal two angles of another, they are similar. Yes: 90° and 35° equal in both ⇒ AA	1+1
16	$(0,y)$: $36+(5-y)^2=16+(3-y)^2 \Rightarrow 61-10y=25-6y \Rightarrow y=9$ ⇒ $(0,9)$	1+1
17	$(x-3)^2+9=25 \Rightarrow x-3=\pm4 \Rightarrow x=7$ or -1	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$d=-2$, $n=\frac{10-34}{-2}+1=13$; $S=\frac{13}{2}(34+10)=286$	1+1+1
19	In $\triangle AOB$ and $\triangle COD$: $\angle AOB=\angle COD$ (v.o.a.), $\angle OAB=\angle OCD$ (alternate) ⇒ $\triangle AOB\sim\triangle COD$ (AA) ⇒ $\frac{AO}{CO}=\frac{BO}{DO}$ ⇒ $\frac{AO}{BO}=\frac{CO}{DO}$	1+1+1
20	$\frac{-k+5}{k+1}=0 \Rightarrow k=5$ ⇒ 5 : 1; $y=\frac{-20-6}{6}=-\frac{13}{3}$ ⇒ $\left(0,-\frac{13}{3}\right)$	2+1
21	$AB=BC=CD=DA=2\sqrt{2}$, $AC=BD=4$ ⇒ **square**	2+1
22	In $\triangle ADC$ and $\triangle BAC$: $\angle ADC=\angle BAC$, $\angle C$ common ⇒ similar (AA) ⇒ $\frac{CA}{CB}=\frac{CD}{CA} \Rightarrow CA^2=CB\cdot CD$	1+1+1
	OR: Similar triangles (AA): $\frac{1.5}{1.6}=\frac{H}{2.4+1.6} \Rightarrow H=\frac{1.5\times4}{1.6}=3.75$ m	1+1+1
23	Mid-point $(0,5)$; quarter points $\left(-1,\frac{7}{2}\right)$, $(0,5)$, $\left(1,\frac{13}{2}\right)$	1+1+1
	OR: $p=\frac{-5+11}{2}=3$; $AB=\sqrt{64+256}=\sqrt{320}=8\sqrt{5}$	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	$2a+5d=12,\ 2a+15d=32$ ⇒ $d=2,\ a=1$; $S_{10}=5(2+18)=100$; $a_{20}=1+38=39$	2+2+1
	OR: (i) $5000+200(n-1)=7000 \Rightarrow n=11$ ⇒ year 2005 (ii) $S_{11}=\frac{11}{2}(5000+7000)=$ ₹66000 (iii) $a_{20}=5000+19\times200=$ ₹8800 ⇒ claim false	2+2+1
25	(i) $\angle A=\angle C$ (opposite angles), $\angle AEB=\angle CBF$ (alternate, $AE\parallel BC$) ⇒ AA (3) (ii) $\frac{AB}{CF}=\frac{AE}{CB} \Rightarrow CF=\frac{8\times9}{12}=6$ cm (2)	3+2
	OR: $\frac{AB}{PQ}=\frac{BC}{QR}=\frac{2BD}{2QM}=\frac{BD}{QM}$ and $\angle B=\angle Q$ ⇒ $\triangle ABD\sim\triangle PQM$ (SAS) ⇒ $\frac{AB}{PQ}=\frac{AD}{PM}$	2+2+1

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ANSWER KEY & MARKING SCHEME — SET C

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** $4n+3$	$7+(n-1)4 = 4n+3$
2	**(b)** Student A is correct	$\frac{20\times21}{2} = 210$
3	**(b)** $-10, -6, -2, 2, \ldots$	Constant difference 4
4	**(a)** $n(n+1)$	$\frac{n}{2}(2+2n) = n(n+1)$
5	**(a)** 8 cm	$\frac{AB}{DE} = \frac{BC}{EF} \Rightarrow EF = 8$
6	**(a)** 10 cm	$\frac{DE}{BC} = \frac{2}{5}$
7	**(d)** AA	Two angles equal ⇒ AA
8	**(a)** similar	All angles 90°, sides proportional
9	**(b)** $(4, 1)$	$\left(\frac{2+6}{2}, \frac{-3+5}{2}\right)$
10	**(c)** ± 4	$9 + p^2 = 25$
11	**(a)** on the negative x-axis	Ordinate 0, abscissa negative
12	**(b)** 2 : 1	$\frac{3k}{k+1} = 2 \Rightarrow k = 2$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$11 - 3(n-1) = -150 \Rightarrow n - 1 = \frac{161}{3}$, not a whole number ⇒ claim is **false**	1+1
14	$\angle DOC = 180^\circ - 125^\circ = 55^\circ$; $\angle DCO = 180^\circ - 70^\circ - 55^\circ = 55^\circ$; $\angle OAB = \angle OCD = 55^\circ$	1+1
15	$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{6}{DB} = \frac{8}{4} \Rightarrow DB = 3$ cm	1+1
16	$AB^2 = 25,\ AC^2 = 25,\ BC^2 = 50$ ⇒ $AB = AC$ and $AB^2 + AC^2 = BC^2$ ⇒ claim is **correct**	1+1
17	$AB = \sqrt{5},\ BC = \sqrt{212},\ AC = \sqrt{265}$; $\sqrt{5} + \sqrt{212} \approx 16.8 \neq 16.3$. Not collinear — claim is **false**.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$a = 200,\ d = 50$: $S_{30} = 15[400 + 29 \times 50] = 15 \times 1850 = ₹27750$	1+1+1
19	$\frac{AD}{AB} = \frac{AE}{AC}$ (since $AD = AE$, $AB = AC$) and $\angle A$ is common ⇒ $\triangle ADE \sim \triangle ABC$ (SAS)	1+1+1
20	Join AC meeting EF at G. In $\triangle ADC$, $EG \parallel DC$ ⇒ $\frac{AE}{ED} = \frac{AG}{GC}$. In $\triangle CAB$, $GF \parallel AB$ ⇒ $\frac{AG}{GC} = \frac{BF}{FC}$. Hence $\frac{AE}{ED} = \frac{BF}{FC}$.	1+1+1
21	$(x-7)^2 + (y-1)^2 = (x-3)^2 + (y-5)^2 \Rightarrow -8x + 8y + 16 = 0 \Rightarrow x - y = 2$	1+1+1
22	Diagonals bisect each other: $\left(\frac{1+x}{2}, 4\right) = \left(\frac{7}{2}, \frac{5+y}{2}\right) \Rightarrow x = 6,\ y = 3$	1+1+1
	OR: $(1:2)$ $\left(\frac{-7+4}{3}, \frac{4-4}{3}\right) = (-1, 0)$; $(2:1)$ $\left(\frac{-14+2}{3}, \frac{8-2}{3}\right) = (-4, 2)$	1½+1½
23	$5 + 3(n-1) = 50 \Rightarrow n = 16$; $S_{16} = 8(5+50) = 440$	1½+1½
	OR: $2a + 10d = 24,\ 2a + 14d = 44 \Rightarrow d = 5,\ a = -13 \Rightarrow -13, -8, -3$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Given, To prove, figure (1). Join BE, CD; draw $EL \perp AB$, $DM \perp AC$. $\frac{ar(ADE)}{ar(BDE)} = \frac{AD}{DB}$, $\frac{ar(ADE)}{ar(DEC)} = \frac{AE}{EC}$ (2). $ar(BDE) = ar(DEC)$ (same base DE, between same parallels) ⇒ $\frac{AD}{DB} = \frac{AE}{EC}$ (2)	1+2+2
	OR: Given $\frac{AD}{DB} = \frac{AE}{EC}$; suppose $DE \not\parallel BC$, draw $DE' \parallel BC$. By BPT $\frac{AD}{DB} = \frac{AE'}{E'C} \Rightarrow \frac{AE}{EC} = \frac{AE'}{E'C} \Rightarrow \frac{AC}{EC} = \frac{AC}{E'C} \Rightarrow EC = E'C$ ⇒ E and E′ coincide ⇒ $DE \parallel BC$	1+2+2
25	(i) 10 km each (ii) $\sqrt{196+4} = 10\sqrt{2}$ km (iii) $(-1, 7)$ (iv) $100 + 100 = 200 = S_1S_2^2$ ⇒ right angle	1+2+1+1
	OR: P $(1:2) = \left(2, -\frac{5}{3}\right)$, Q $(2:1) = \left(0, -\frac{7}{3}\right)$ (3). $AP = \sqrt{4 + \frac{4}{9}} = \frac{2\sqrt{10}}{3} = PQ$ ✓ (2)	3+2

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